

Prarochana AI: AI-Enabled Ad Campaign Generation and Analysis

Chittesh Kumar Kanipakam	Dhruv Nilkund	Dhruv Poduval	Ganta Srilalitha	Dr. Chaitra N
<i>Dept. of CSE</i>	<i>Dept. of CSE</i>	<i>Dept. of CSE</i>	<i>Dept. of CSE</i>	<i>Dept. of CSE</i>
<i>PES University</i>	<i>PES University</i>	<i>PES University</i>	<i>PES University</i>	<i>PES University</i>
Bengaluru, India	Bengaluru, India	Bengaluru, India	Bengaluru, India	Bengaluru, India
chitteshkumar13@gmail.com	dnilkund@gmail.com	dhruvpoduval@gmail.com	srilalitha2003@gmail.com	chaitran@pes.edu

Abstract—Digital marketing, in today’s digital era, is an evolving trend in the marketing sector. It boosts businesses to reach a wider audience, including other businesses and customers through appropriate advertising and effective marketing strategies for their products. The traditional marketing methods incur high costs, time, third-party agencies, and a group of expert analysts responsible for drafting the ads, including inefficiency in conveying an appropriate message to the target audience, and difficulties in managing multiple campaigns. To overcome these challenges, we present Prarochana AI, an innovative application of Generative AI in automating advertisement campaign generation tailored to the beverage industry. Leveraging the dataset containing ad campaigns of the beverage industry, the application facilitates the creation of engaging and effective ad content using LLMs, RAG LLMs, and Image generation models. Besides, it generates an approximate and optimized budget that must be allocated for the campaign across various digital platforms, ultimately enhancing marketing efficiency and consumer engagement. Our proposed application enables users to get meaningful content for advertising, and campaign plans and have a seamless experience.

Index Terms—Generative AI, LLMs, RAG LLMs, Advertising automation, Beverage industry, Image Generation models, Dataset utilization, campaign generation

I. INTRODUCTION

A. Background

In this digital era, creating effective ad campaigns is crucial for brands to enhance their market presence. Traditional methods for launching and maintaining a campaign over time for brands and their products incur high costs, time, and labor of expert analysts for buying media platforms and paying third-party agencies for drafting and shooting the ad. A campaign can fail due to improper targeting, improper allocation of resources to various channels in a media mix, improper message conveyed to the target audience, and lack of analysts responsible for tracking the campaign’s performance. Generative AI offers solutions to these challenges. Our application “Prarochana AI” is leveraged on a dataset containing Indian Beverage Advertisements, providing customers with meaningful ad content that aligns with their needs and ad creatives that include images, along with a budget allocation plan, that recommend them the approximate budget and resources to be allocated on different channels for their ads.

B. The evolution of the Advertising Domain through Gen AI

Generative AI (Gen AI) in the marketing and advertising sectors has led to an evolution enabling the generation of dynamic, personalized campaigns that align with user preferences. The advancements of large language models (LLMs), RAG models, and text-to-image models automate ad generation by generating rich and customized content for the specific target audience. Thus, brands can focus more on strategy and less on manual execution, paving the way for a more robust and responsive marketing landscape.

II. LITERATURE REVIEW

A. Text Generation Techniques using LLMs

GPT2[1] employs unsupervised learning and modified layer normalization to generate the next subsequent word. LLaMA[2] uses a streamlined architecture with models trained on fewer parameters to generate relevant text. Mistral 7B [3] employs grouped-query attention (GQA) and sliding window attention (SWA) mechanisms for generating longer contexts.

B. Text-to-Image Generation Techniques

CLIP[4] uses cosine similarity between the image and its respective text prompt embeddings, which provides a measure of relevance between the image and text. Dall-e[5] employs shared embedding space with CLIP and decodes and generates new visuals.

C. Multi-Channel Campaign Generation and Budgeting Techniques

RAG LLMs with vector search[6] and keyword search [9] discuss how LLMs and database systems can be used to fetch relevant documents from databases using **Vector Indexing** and **Keyword Indexing**. Match2One -AI [13] platform, analyzes traffic over websites where the campaigns are hosted and optimizes the traffic by finding the right target audience for generating awareness for their brand.

III. METHODOLOGY AND APPROACH

The approach for generating a full fledged ad campaign involves a sequence of steps as described in Fig 1:

- 1) Collecting, pre-processing and storing of AdBits Dataset (Section IV)

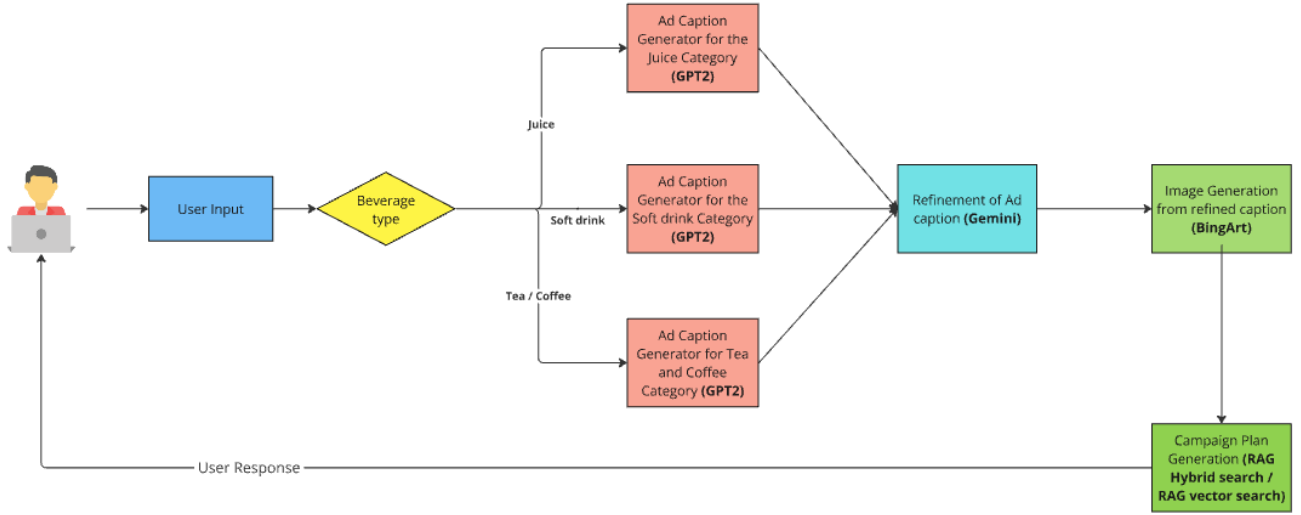


Fig. 1. End-to-End Workflow for a Beverage Ad Campaign Generation

```

gpt2_ad_caption: [COMPANY] Thims Cuurent [MESSAGE] encourages consumers to feel energetic by drinking chilled Thims Current,known for its strong refreshing taste and rising fizz, encouraging consumers to become energetic in life. [TYPE] Soft Drink [DESCRIPTION] carbonated cold drink,strong taste,more fizz [PRICE]Rs 58/750ml /500ml [TARGET AUDIENCE]youngsters,genZ [TV_ad_caption][The video begins with Ranveer Singh entering a bustling city street filled almost entirely with energy. He effortlessly executes daring stunts and dance moves while holding up a can of ThunderBliss on his mobile phone. As he takes breaks between sets of commands—a quick sip of the refreshing beverage invigorates him, allowing him to continue executing thrilling maneuvers amidst crowds—he seamlessly transitions into scenes of everyday joy, from upbeat music to spontaneous dancing. The dynamic visuals mirror those of urban living areas, enhancing the sense that moments are alive when shared among people of all ages. With each impressive move or grab Daniel grabs a Sprite, the scene shifts to festive festivities unfolding along with lively camaraderie. Amidst vibrant celebrations, Salman leaps onto the final_refined_ad_caption: A whirlwind of energy and charisma, a dynamic performer in a vibrant outfit, navigates a bustling cityscape. Amidst daring stunts and electrifying dance moves, he pauses for a refreshing sip of Thims Cuurent, rejuvenated to continue his energetic display. The vibrant backdrop reflects the joy and camaraderie of everyday life, underscoring the power of shared moments.
  
```

Fig. 2. gemini and gpt2 generated captions

- 2) Collection of user inputs as defined in Section VIII.
- 3) Use of Text generation models to generate ad caption from user inputs (Section V)
- 4) Use of Image generation models to generate ad images from ad caption generated (Section VI)
- 5) Budget Models using RAG for campaign generation based on user inputs (Section VII)

IV. DATASET USED

The AdBits dataset contains a wide array of Indian Beverage Advertisement campaigns. The dataset was created by aggregating advertisements from diverse sources, including digital platforms, television, and print media. It includes critical attributes such as the name of the campaign, year, message conveyed, product description, price, brand ambassadors and their personality traits, target audience, slogans, competitor brands, success / failure of the campaign and a reason to follow it with an approximate budget (using statistical equations like Engagement Rate) spent on various channels such as Facebook, print, television, etc. This data is crucial for training different models (Section V) and the generation of ad creatives.

V. GENERATING THE ADVERTISEMENT CAPTIONS FOR EACH CHANNEL

A. Generating the ad-captions using Mistral 7-B

Mistral 7B[3], an open-source model is known for its high-quality text generation. Mistral 7B has a better ability

to capture context which helped in generating ad captions that aligned with the user input. The **training loss (Cross Entropy Loss)** and **ROUGE-score** obtained was **0.57 ,0.52** respectively. However, the model in its training phase showed that it required higher computational resources and higher latency leading to higher response times. The complexity in fine-tuning the model required us to adapt memory-efficient methods like **LoRA (Low-rank adaptation)**, hence limiting the quality of its generated ad captions when deployed on a bigger scale. Due to overfitting, the captions generated by Mistral 7B, were highly repetitive, overly specific, and did not generalize well for new user inputs and new product descriptions.

B. Generating the ad-captions using Llama 7B

Llama (Large Language Model Meta AI) [2] 7B model was trained to generate ad captions given the user input. This model is lighter and computationally more efficient as compared to Mistral 7B. The **training loss** and **ROUGE-score** obtained were **0.843 ,0.43** respectively. But the model lacked in generating ad captions that aligned with the target audience in the user input. The generated ad captions looked repetitive and duplicated.

C. Generating the ad-captions using GPT-2

GPT-2 model was trained to generate ad captions given the user input. With its power to generate creative ad captions and

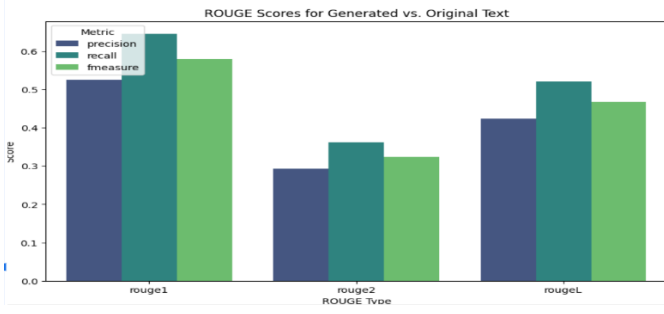


Fig. 3. ROUGE score for GPT2

the ease of training, it outperformed Mistral 7B and Llama 7B. The generated ad captions were unique and highly aligned with the target audience and product description as given by the user. Computational costs were minimal while training GPT-2 by using its open-source version provided by HuggingFace (GPT2LMHeadModel) which helped track its performance. The **training loss** and **ROUGE-score** obtained were **0.406** ,**0.57** respectively. It is the best ROUGE score obtained so far, outperforming Mistral 7B and Llama 7B (Fig 3).

D. Using Gemini to replace brand ambassador with characteristic traits

Gemini 1.5 Flash is used to replace the brand ambassador's name present in the caption generated by GPT2 with its characteristic features and personality traits. Not doing so would lead to intrusion of their privacy. The refined ad caption can be seen in Fig 2.

VI. GENERATING ADVERTISEMENT IMAGE FROM AD CAPTION PROMPT

A. Using Dall-E for generating images

The generated ad-caption is then fed to generate the caption and customer specific ad-image. As seen in Section IV, Dall-e 3 produces high quality images and outperforms the stable diffusion models, as justified by the following metrics-

- Clip Score [4]:** cosine similarity score between image and text embeddings. The Clip (Contrastive Language-Image Pretraining) score model has not been trained on publicly available annotated image datasets containing specific beverage bottles, brand ambassadors, or water text inscriptions. Hence, an average clip score of **44.301** is generated.
- Inception Score:** measures similarity with real image and diversity of generated images. Since the generated images contain a persona and not an existing individual with a diverse backdrop, an inception score of **1.00** is obtained.
- Human Evaluation:** human judgment of the quality of the generated images.

B. Using BingArt for generating images

BingArt Dall-e generates a set of 4 images for a given ad-caption. Each image is unique and aligns with the ad-caption (Fig 7). These images are recommended to user for

advertising their beverage across multiple digital platforms such as Facebook, Twitter, YouTube and Instagram.

VII. BUDGET PREDICTION USING RAG MODEL

Retrieval Augmented Generation (RAG) is a workflow of LLMs (Large Language Models) that augments the result of information retrieval systems with LLMs to generate a user-friendly and human-understandable response.

A. RAG using Hybrid search

Hybrid search is a fusion of **Vector Search** and **Keyword Search**. Vector Search includes use of **dense vectors** and **cosine similarity** to retrieve **semantically similar** documents from the dataset. Keyword Search includes use of **sparse vectors** and **dot product** to retrieve **syntactically similar documents**. The dataset (Section IV) which is stored and preprocessed, along with *user queries* is converted to dense vector embeddings using **all-MiniLM-6v2** [11] and sparse vector embeddings using **BM25Encoder** [10] embedding models. These embeddings are stored in the **PineCone** Database. **Top Results₁(k)** and **Top Results₂(k)** are the top k results obtained from vector search and hybrid search respectively, as seen in Fig 4. **Rank=[1,k]**, k is the no. of documents. A default weight of **0.5** is assigned, implying that each k-th result obtained from both search strategies is considered equally.

The score of each k-th document is given by:

$$R1_k = \frac{1}{\text{Rank of } k\text{-th document in Results}_1(k)} \quad (1)$$

$$R2_k = \frac{1}{\text{Rank of } k\text{-th document in Results}_2(k)} \quad (2)$$

$$\text{Score}_k = R1_k + R2_k \quad (3)$$

This is known as **Reciprocal Rank Fusion (RFF)** [14], a metric that merges the results of two search strategies. The final result would be a list of k documents arranged in decreasing order of their *RFF* scores which is then augmented by the LLM **PineconeHybridSearchRetriever** [12] to generate a user-friendly response to the user's query. Fig 8 shows the approximate budget generated using hybrid search with a score of **0.45**.

B. RAG using Vector search

To estimate the budget for a user's ad campaign, we used vector search to find and analyze similar previous campaigns captured in the dataset (Section IV). The preprocessed dataset and user's query are converted to dense vectors using **General Text Embedding (GTE-Large)** [8] embedding model. These embeddings are stored in a shared cluster in a vector database named **Mongo Atlas**. **Vector search** is employed to retrieve a set of documents that are the most semantically similar (determined by **cosine similarity**) to the user's query. **Gemma-2b** [7] is the LLM used to augment the results and returns a human-readable response to the user as seen in Fig 5. Fig 9 shows the approximate budget generated across various channels with a maximum similarity score of **0.9287**.

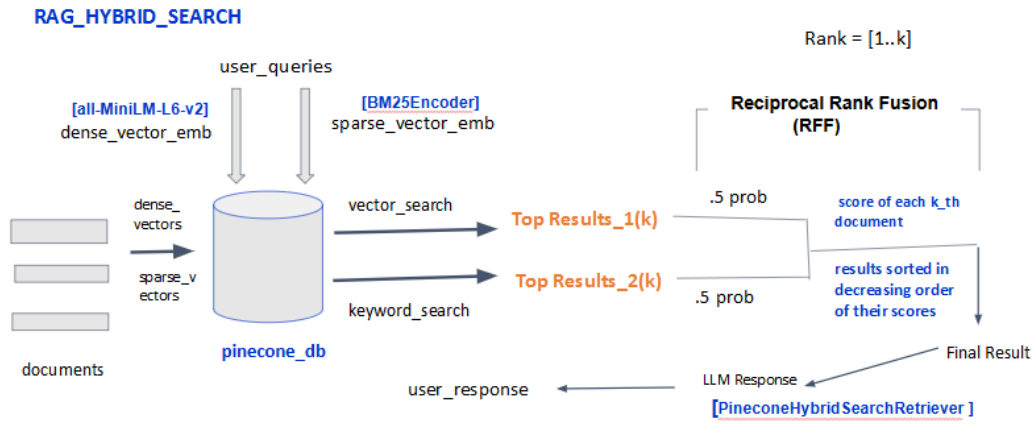


Fig. 4. Hybrid Search Pipeline.

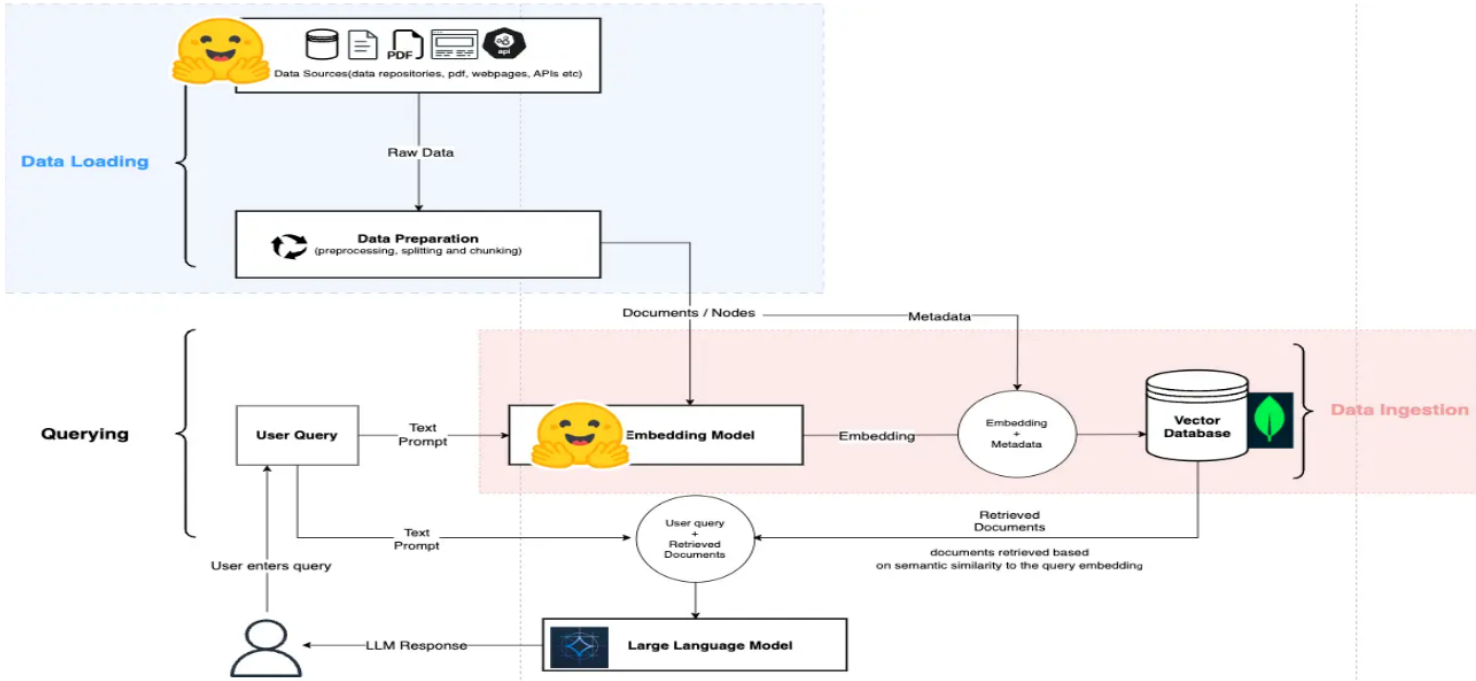


Fig. 5. Vector Search Pipeline.

VIII. PSEUDOCODE OF THE APPLICATION

Some annotations:

- **bev_type**: Beverage type (e.g., Juice, Soft Drink, Tea/Coffee)
- **msg**: Message to be conveyed in the advertisement
- **input**: A set of input parameters, including **company_name**, **bev_type**, **price**, **msg**, **description**, and **target_audience**
- **cad**: cleaned-ad-caption

```
def generate_ad(input):
    if bev_type == 'Juice':
        return generate_ad_for_juice(input)
    elif bev_type == 'Soft Drink':
        return (
```

```
generate_ad_for_softdrink(input))
elif bev_type == 'Tea/Coffee':
    return (
        generate_ad_for_tea_coffee(input))
```

```
def generate_ad_caption(input):
    #GPT_2 model trained on specific input
    bev_type to return an ad_caption.
    return GPT_2(input)
```

```
def clean_ad_caption(ad_caption):
    #Gemini model to replace ambassador's
    name with their personality traits, to
    generate a persona.
```

```

return Gemini(ad_caption)

def generate_ad_image(cad):
    #returns a set of 4 images,
    generating a persona with diversified
    backdrop setting.
    return BingArt(cleaned_ad_caption)

def generate_budget(input):
    # RAG_HYBRID_SEARCH / RAG_VECTOR_SEARCH
    return Budget_Model(input)

# All bev_types share the below common
#workflow
def generate_ad_for_juice(input):
    ad_caption = generate_ad_caption(input)
    cad = clean_ad_caption(ad_caption)
    ad_img = generate_ad_image(cad)
    budget = generate_budget(input)
    return ad_img, budget

```

IX. FINAL MODELS CHOSEN FOR THE APPLICATION

The following models were finally chosen for our Prarochana AI application:-

- For ad caption generation: GPT-2
- For ad image generation from ad caption: BingArt Dall-e
- For budget and campaign plan generation: RAG using Hybrid Search and RAG using Vector Search

X. DEVELOPING A GUI FOR A FULLY FUNCTIONAL PRAROCHANA AI APPLICATION

The UI (User Interface) framework chosen for our application is Streamlit. It is an open-source Python library that enables efficient development of interactive web applications and eases the integration of machine learning models.

A. Streamlit Integration with Prarochana AI

Streamlit bridges the gap between the back end that contains various AI models that get triggered once they receive input from the user through the front end. The front end features a user-friendly form asking users the details of the beverage they want to advertise. After filling out the form and clicking on the "Generate Ad Campaign" button, the application displays custom-generated ad images along with a comprehensive budget plan, outlining recommended spend across various social media platforms. As illustrated in Fig 1, the workflow of generating an ad campaign is structured as follows:

- 1) The application presents the user with a form to enter their preferences for generating an ad campaign.
- 2) Once, the form is filled, based on the beverage type, the Ad Caption Generator module triggers the GPT2 model that was trained for that specific beverage type (Fig 6).
- 3) Generated ad captions by GPT2 are further enhanced using the Gemini model.
- 4) The refined ad captions, are then passed to the BingArt Dall-e model to generate corresponding ad images.

Ad Generation - Beverage Campaign

Enter the company name:

Nature's Own Delight

Enter the Message to be conveyed to target audience:

Experience the irresistible taste and refreshment of Nature's Own Delight, made with 100% pure fruit juice.

Enter the beverage type:

Juice

Enter the Product description:

Nature's Own Delight is made with the finest fruits, carefully selected for their sweetness and flavor. Each variant captures the unique taste and aroma of the fruit it is made from, providing a burst of juicy goodness in every sip.

Enter the price of the product per 500 ml:

Rs 35

Enter the Target audience:

Families, Young adults, Health-conscious individuals.

Generate Ad Images

Fig. 6. User input given in the UI

- 5) Finally, the application provides a campaign plan, outlining the approximated budget model for various social media platforms (Fig 7, Fig 8 and Fig 9).



Fig. 7. Generated Ad images displayed in our UI

XI. TESTING AND RESULTS OBTAINED

The objective of testing this pipeline was to ensure that the generated ad captions and the ad images, were qualitatively and quantitatively good. **ROUGE (Recall-Oriented Understudy for Gisting Evaluation score) scores**, an open-source evaluation metric was used to test the quality of the generated captions against ground truth(original captions) which can be calculated using hugging face's evaluate library. The ad images were evaluated using **CLIP score** (Section VI A).

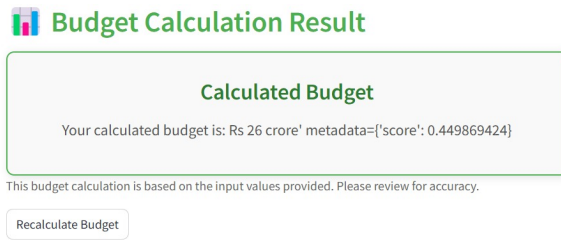


Fig. 8. Budget calculated using RAG Hybrid Search being displayed in UI

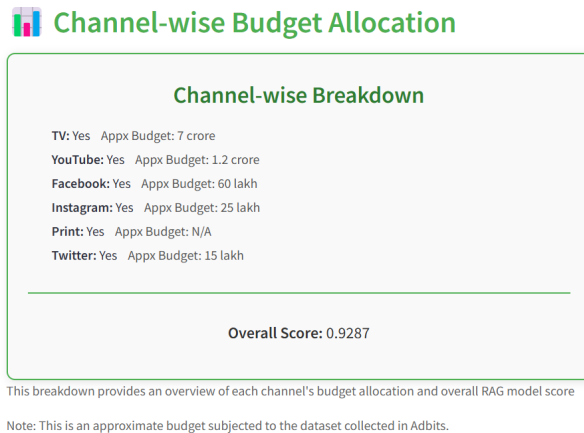


Fig. 9. Channel wise budget allocation using RAG Vector Search

Integration testing was carried out to validate the combined performance of the pipeline, from ad captions to ad images. The application was tested with various user inputs, and the pipeline ran successfully and generated appropriate ad images and budget outputs for the campaigns, for all the inputs. **Non-functional testing** included both usability and performance testing. In **usability testing**, Google Forms were used to gather user feedback on our Prarochana AI application. The survey was conducted on a sample population of size 80, where they were asked to choose between our AI generated ad images in comparison to the real ad images. The responses revealed that 73 percent of the audience found our AI generated ad images more appealing and attractive, confirming our model's ability to produce ad images that resonated with a broader audience (Fig 10). **Performance testing** was also conducted to assess the models' response times, measuring the speed of generated ad captions and ad images. The results demonstrated that our models' were able to deliver content in lesser response time and hence ensured efficiency.

XII. FUTURE CHALLENGES AND SCOPE

The Prarochana AI application brings numerous advantages to the advertising landscape, particularly in the beverage industry. The application is a powerful tool for brands to generate impactful advertisements without significant expenses or intensive manual effort. The application can then be expanded beyond beverages to other sectors within the fast-moving consumer goods (FMCG) industry. Further enhancements could

Q1) Which among the Beverage juice ad do you prefer?

79 responses

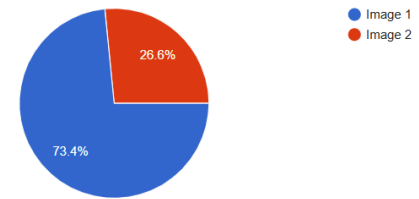


Fig. 10. Google form responses for AI Fruit Juice ads(Image1) and corresponding real life ad for the Juice(Image2)

include providing recommendations for optimal ad placement timing, such as identifying high-traffic periods (e.g., prime time) on various media channels. By using AI to streamline campaign creation, brands can focus on strategic initiatives while ensuring their ads are relevant, engaging, and tailored to their target audience.

REFERENCES

- [1] A. Radford, J. Wu, R. Child, D. Luan, D. Amodei, and I. Sutskever, "Language Models are Unsupervised Multitask Learners," OpenAI, 2019.
- [2] H. Touvron, T. Lavril, G. Izacard, E. Grave, P. Preux, B. Martinez, A. Bachoc, M. Lachaux, T. Goyal, E. Nie, P. J. He, M. Ott, F. Hernandez, L. Joulin, E. Hazan, and J. Copet, "LLaMA: Open and Efficient Foundation Language Models", arXiv:2302.13971, 2023.
- [3] A. Q. Jiang, A. Sablayrolles, A. Mensch, C. Bamford, D. S. Chaplot, D. de las Casas, F. Bressand, G. Lengyel, G. Lample, L. Saulnier, L. R. Lavaud, M.-A. Lachaux, P. Stock, T. Le Scao, T. Lavril, T. Wang, T. Lacroix, W. El Sayed, "Mistral 7B," in NeurIPS, 2023.
- [4] J. Hessel, A. Holtzman, M. Forbes, R. Le Bras, Y. Choi, "CLIPScore: A Reference-free Evaluation Metric for Image Captioning," 2021.
- [5] Ramesh, A., Pavlov, M., Goh, G., Gray, S., Voss, C., Radford, A., Chen, M., Sutskever, I. "Zero-shot text-to-image generation." Proceedings of the 38th International Conference on Machine Learning (ICML), 2021.
- [6] MongoDB Inc. "Gemma: Integrating MongoDB Atlas with Hugging Face's RAG for Advanced AI Applications," MongoDB Developer Hub. [Online]. Available: <https://www.mongodb.com/developer/products/atlas/gemma-mongodb-huggingface-rag/>.
- [7] Gemma Team, Google DeepMind, Gemma 2: Improving Open Language Models at a Practical Size," arXiv preprint arXiv:2408.00118, 2024.
- [8] Z. Li, X. Zhang, Y. Zhang, D. Long, P. Xie, and M. Zhang, "Towards General Text Embeddings with Multi-stage Contrastive Learning," arXiv preprint arXiv:2308.03281, 2023.
- [9] Krish Naik, Hybrid Search RAG With Langchain And Pinecone Vector DB, YouTube, [Video]. Available: <https://www.youtube.com/watch?v=CK0ExcCWDp4>.
- [10] Milvus, Embedding with BM25, Milvus Documentation. Available: <https://milvus.io/docs/embed-with-bm25.md>.
- [11] Hugging Face, sentence-transformers/all-MiniLM-L6-v2, Hugging Face. Available: <https://huggingface.co/sentence-transformers/all-MiniLM-L6-v2>.
- [12] LangChain, "Pinecone Hybrid Search Integration," LangChain Documentation. Available: https://python.langchain.com/docs/integrations/retrievers/pinecone_hybrid_search/.
- [13] O. Rask, "What is Programmatic Advertising? The Ultimate 2020 Guide." June 23rd, 2020. Available: <https://www.match2one.com/blog/what-is-programmatic-advertising/>.
- [14] G. V. Cormack, C. L. A. Clarke, and S. Buettcher. 2009. Reciprocal rank fusion outperforms Condorcet and individual rank learning methods. In Proc. ACM Conf. on Research and Development in Information Retrieval (SIGIR). 758–759.